
Pursuing a **minor** in Machine Intelligence & Data Science at CMinDS, IIT Bombay
Cleared **CFA® Level I** - August 2024

SCHOLASTIC ACHIEVEMENTS

- Selected for a **fully-funded** semester exchange to **EPFL, Switzerland** (1 out of 3 students from IITB) [*'25*]
- Conferred with **Undergraduate Research Award** (URA-01) by IITB for exceptional undergraduate research [*'24*]
- Awarded the **NUS Young Fellow Scholarship, Singapore** for GenAI conference (only 6 students from IITB) [*'24*]
- Nominated for **AUA-THU Global School** by **Tsinghua University, China** (1 out of 2 students from IITB) [*'23*]
- Secured an **AP** grade in the course DS303 (Introduction to Machine Learning), awarded to 1 out of **173** students [*'23*]
- Secured **99.3** percentile in Joint Entrance Examination (**JEE**) **Advanced** and **99.78** percentile in **JEE Mains** [*'21*]

WORK EXPERIENCE

Investment Research Analyst | Axis Asset Management Corporation (AMC) [*May '24 - Jul '24*]
Awarded a **PPO** to work with the quantitative strategy division managing quant funds with **AUM INR 1000+ Cr**

- Engineered a Python model to **predict NIFTY 100 index reconstitution** to assist an **active quant fund** saving **10+** recurring operation hours in stock selection monthly & **improved tracking** of passive money to **>90%** accuracy
- Backtested algorithms with **>93%** accuracy to forecast **5** securities that will enter/exit the index at any given time
- Conducted a **historical** analysis of **NIFTY 500** companies, examining **quarterly** variations on multiple bases for large, mid and small cap segments for last **10** years, tracking **sectoral transitions** and **company segment shifts**

Financial Research Analyst | National Stock Exchange (NSE) of India [*Nov '23 - Jan '24*]
Received a **letter of recommendation** from the **Chief Economist** of NSE for active contribution to the **EPR** team

- Conducted an in-depth analysis of key changes over years **validating** the **hypothesis of financial illiteracy** among new traders in FO segment & compiled a report '**Pre and Post-COVID trading behaviour of retail investors**'
- Analysed over **240** million tick-by-tick order data points leveraging **R** to model PnL statements of **5+** million traders
- Played a key role in the development of ongoing projects highlighted in NSE's monthly '**Market Pulse**' publication

KEY PROJECTS

**Supervised Learning Projects (SLPs) involve semester-long R&D under an IIT-B professor, equivalent to a full 6-credit course*

Modeling Intraday Returns and Volatility using ARIMA-GARCH models [*Autumn '25*]
B.Tech Project | Guide: Prof. Piyush Pandey, SJM School of Management, IIT Bombay

- Conducted **financial analysis & forecasting**: utilizing **ARIMA-GARCH** models for CNXBAN index prediction
- Leveraged **second-by-second financial data** to analyse **volatility clustering**, periodic intraday patterns in **CNXBAN** and integrating **ARIMA** models with **GJR-GARCH (1,1)** for informed high-frequency forecasting

Adoption of EV Usage in Rural areas : Barriers and Challenges [*Spring '24*]
Supervised Study, secured an **IRB** approval for further research | Guide: Prof. Anand Rao, CTARA, IIT Bombay

- Conducted **10+** EV expert surveys, focusing on **9** major barriers and sub-barriers to **adoption of EVs** in **3** villages
- Leveraged the **Fuzzy DEMATEL** model to measure **15+** parameters from the survey data for ranking the barriers
- Mathematically derived a **cause-effect graph** among barriers to **bridge gap** between **expert and public opinions**

Semantic Table, Column & Cell recognition using TableNet [*Autumn '24*]
Supervised Learning Project | Guide: Prof. Biplab Banerjee, CSRE Department, IIT Bombay

- Managed a **Swiss blockchain** client to develop a ML-based image segmentation model, delivering **>90%** accuracy
- Developed robust models for table extraction with **TableNet** & **DenseNet** pipelines, requiring **10×** lesser input data

TECHNICAL SKILLS

Languages	: C++, Python, R, HTML, CSS, JS, SQL, Bootstrap, ROOT
Data Science	: NumPy, Pandas, Matplotlib, Seaborn, MATLAB, GNU Octave
Softwares	: LTSpice, MSOffice, Bloomberg

EXTRACURRICULARS

- Secured a funding of **500K+ INR** for a startup pitched in the level 2 of **IDEAS** cohort by **DSSE, IITB** [*Mar '24*]
- Part of the **Inter IIT 2024** Cricket team, with only **15** selections, and won a **Gold** medal among **23 IITs** [*Dec '24*]
- Awarded the '**Most Valuable Player**' and '**Best Bowler**' in Inter-Hostel Cricket Championship 2023-24 [*Jan '24*]
- Participated and **won** the Aavhan's Revive Sports League (**RSL**) in Cricket out of **6** teams institute-wide [*Apr '22*]
- Completed a yearlong training program in National Sports Organisation (**NSO**) in **Cricket** at IIT Bombay [*Dec '22*]
- Achieved **10,000+ INR** sales with **self-published** book on CSS proficiency enhancement on **Gumroad** [*Jun '22*]